VIP Individual Project Part 2: Beam Design and AI Model Comparison

Experiment status: All 44 AI model proposals completed across four live runs. All performance metrics were obtained using the unmodified instructor-supplied `beam_model.py` (dimensions in mm).

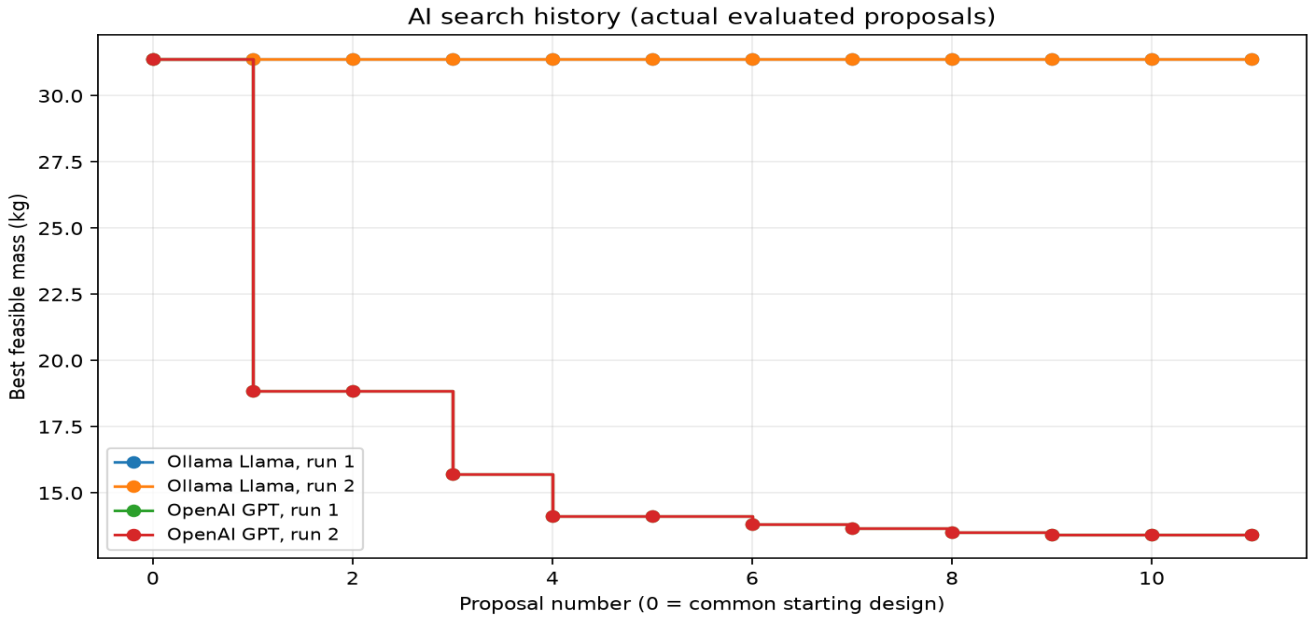
Results comparison

Method / model	Run	Best mass (kg)	Width (mm)	Height (mm)	Defl. margin (mm)	Stress margin (MPa)	Evals	Invalid replies	Time (s)
SciPy SLSQP scipy.optimize.minimize	1	13.4233	20.000	85.499	0.0000	17.401	152	N/A	0.01
Ollama Llama llama3.2:1b	1	31.4000	50.000	80.000	0.5117	73.125	12	0	45.07
Ollama Llama llama3.2:1b	2	31.4000	50.000	80.000	0.5117	73.125	12	0	29.32
OpenAI GPT gpt-4.1-mini	1	13.4235	20.000	85.500	0.0000	17.404	12	0	10.64
OpenAI GPT gpt-4.1-mini	2	13.4235	20.000	85.500	0.0000	17.404	12	0	9.09

SciPy convergence and evaluation accounting

SciPy SLSQP converged successfully: **Optimization terminated successfully.** The CSV logs 152 calls to the common beam evaluator (starting design, objective and constraint calls, final result, and 10 near-boundary floating-point feasibility checks). These 10 checks are **not invalid AI replies**. The optimized design is 20.0000 × 85.4988 mm; mass 13.4233 kg, deflection approximately 1.0000 mm, and stress 102.5986 MPa.

Search history: best feasible mass



The curves track the best feasible mass after each evaluated proposal, not the mass of every individual candidate.

Interpretation of AI behavior

Both GPT runs reduced the best feasible mass from 31.4 kg to 13.4235 kg at proposal 9 (20 × 85.5 mm), nearly matching the SciPy result. The GPT runs followed the same proposal sequence and each tested infeasible lighter candidates (for example, 20 × 80 mm and 20 × 85.4 mm), but only feasible designs were eligible for the best-mass record. This pattern is consistent with using deflection feedback

to narrow toward the boundary.

Both Llama runs remained at the initial best feasible mass of 31.4 kg. Llama consistently proposed heavier designs, including repeated 90 × 100 mm designs in run 1. Run 2 varied some later proposed dimensions, but neither run improved the starting design. All 44 model replies were valid JSON proposals within bounds; 0 invalid replies occurred. Variation in proposed dimensions did not lead to variation in the final best mass for either model.

Limitations and reproducibility

The shared evaluator is idealized small-deflection beam theory, not a 3D finite-element model; it omits self-weight, local load/support effects, and impacts. The comparison uses different evaluation budgets (152 SciPy calls versus 12 per AI run), so it does not isolate sample efficiency. Only two runs per model were conducted; outcomes depend on prompt, sampling settings, and software versions. Models: **llama3.2:1b** via Ollama 0.40.1, **gpt-4.1-mini** via OpenAI API; temperature 0.2; common starting design 50 × 80 mm; 11 proposals per run; strict JSON response validation; no automatic repair of invalid proposals.

GitHub repository: https://github.com/sais03687/VIP-Beam-AI-Model-Comparison